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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	G.Mamatha
Student Name	Penmatsa Gowtham Varma
Roll Number	2520090202
Branch	CSIT
Course Outcome	CO3

Case Study Topic: Mumbai Distribution Grid – Prim's MST and Bridge Detection for N-1 Redundancy

Work Done Summary:
In this case study, a power distribution network was modeled using graph algorithms. Prim's Algorithm was used to construct the Minimum Spanning Tree, and Bridge Detection was performed to identify critical connections and improve network reliability.

Details:

• Created a weighted graph representing substations and cable connections.

• Implemented Prim's Algorithm using a Priority Queue.

• Generated the Minimum Spanning Tree (MST).

• Calculated the total construction cost of the MST.

• Implemented DFS-based Bridge Detection.

• Identified critical bridge edges in the network.

• Added augmentation edges to create cycles.

- Eliminated bridge edges to achieve N-1 redundancy.

Result: Screenshots/Output

Output

```
MUMBAI DISTRIBUTION GRID
------------------------
MST Cost = 20 Crores
Additional Edge Cost = 5 Crores
Total Network Cost = 25 Crores

N-1 Redundancy Achieved

=== Code Execution Successful ===
```

Observations:

- The graph was successfully represented using adjacency lists.
- Prim's Algorithm generated the minimum-cost spanning network.
- Bridge edges were correctly identified using DFS traversal.
- Additional edges reduced network vulnerability.
- The final network satisfied redundancy requirements.
- System reliability improved without excessive cost increase.

GitHub Link: https://github.com/gowthampenmatsa07-prog/D-S-A.git

Student Signature	Faculty Signature